```
(base) PS E:\CEDT\Hardware Design and Implementation\inclass6> & "..\inclass5\iot-rz\test\sim21.exe" ".\mos-rz2-s.obj"
>b 492
set breakpoint 492
>g
interrupt0
>r

r0 :0    r1 :0    r2 :0    r3 :0    r4 :0    r5 :0    r6 :0    r7 :0
r8 :0    r9 :0    r10:0    r11:0    r12:0    r13:0    r14:0    r15:0
r16:0    r17:0    r18:0    r19:0    r20:0    r21:10148 r22:0    r23:0
r24:0    r25:0    r26:0    r27:0    r28:0    r29:10164 r30:10228 r31:0
PC:492 RetAds:492 timer0 149 timer1 199 clock 599
>d 1100 9
1100 : 10000 10292 2 10148 0 10294 10298 10302 0
>d 10000 2
10000 : 10146 10292
>d 10146 2
10146 : 10002 10292
>d 10292 2
10292 : 10148 10146
>d 10002 6
10002 : 461 10034 10082 0 0 1
>d 10148 6
10148 : 492 10180 10228 0 0 1
>d 10294 12
10294 : 0 10296 0 0 1 10300 0 0 1 10304
10304 : 0 0
>b 250
set breakpoint 250
>g
```

```
interrupt0
>r

r0 :0    r1 :0    r2 :10322 r3 :10300 r4 :1    r5 :0    r6 :0    r7 :0
r8 :0    r9 :0    r10:0    r11:0    r12:0    r13:0    r14:0    r15:0
r16:0    r17:0    r18:0    r19:0    r20:344  r21:10148 r22:0    r23:0
r24:0    r25:0    r26:0    r27:341  r28:10322 r29:10180 r30:10239 r31:0
PC:250 RetAds:0 timer0 2 timer1 52 clock 852
>d 1100 9
1100 : 10000 10292 1 10148 0 10294 10298 10302 1
>d 10000 2
10000 : 10146 10292
>d 10146 2
10146 : 10002 10292
>d 10292 2
10292 : 10148 10146
>d 10002 6
10002 : 461 10034 10082 0 0 1
>d 10148 6
10148 : 344 10180 10239 341 10322 0
>d 10294 12
10294 : 1 10296 0 0 -1 10300 10322 10322 1 10304
10304 : 0 0
>t
PC 250 ld r1 1102        r0:0 r1:1 r2:10322 r3:10300 r4:1 r5:0 r6:0 r7:0 r8:0 r9:0
                    r27:341 r28:10322 r29:10180 r30:10239 r31:0
>g
interrupt0
>d 1108 1
1108 : 1
>r
```

```
r0 :0    r1 :0    r2 :0    r3 :0    r4 :0    r5 :0    r6 :0    r7 :0
r8 :0    r9 :0    r10:0    r11:0    r12:0    r13:0    r14:0    r15:0
r16:0    r17:0    r18:0    r19:0    r20:461  r21:10002 r22:0    r23:0
r24:0    r25:0    r26:0    r27:0    r28:0    r29:10034 r30:10082 r31:0
PC:250 RetAds:0 timer0 64 timer1 114 clock 914
>d 1100 9
1100 : 10000 10146 1 10002 0 10294 10298 10302 1
>d 10000 2
10000 : 10146 10292
>d 10146 2
10146 : 10002 10292
>d 10292 2
10292 : 10148 10146
>d 10002 6
10002 : 461 10034 10082 0 0 1
>d 10148 6
10148 : 344 10180 10239 341 10322 0
>d 10294 12
10294 : 1 10296 0 0 -1 10300 10322 10322 1 10304
10304 : 0 0
>t
PC 250 ld r1 1102        r0:0 r1:1 r2:0 r3:0 r4:0 r5:0 r6:0 r7:0 r8:0 r9:0
                    r27:0 r28:0 r29:10034 r30:10082 r31:0
>g
+1 interrupt0
>d 1108 1
1108 : 1
>r

r0 :0    r1 :0    r2 :10322 r3 :10296 r4 :1    r5 :0    r6 :0    r7 :0
```

```
r8 :0    r9 :0    r10:0    r11:0    r12:0    r13:0    r14:0    r15:0
r16:0    r17:0    r18:0    r19:0    r20:343  r21:10002 r22:0    r23:0
r24:0    r25:0    r26:0    r27:341  r28:10322 r29:10034 r30:10092 r31:0
PC:250 RetAds:0 timer0 77 timer1 127 clock 1327
>d 1100 9
1100 : 10000 10146 1 10002 0 10294 10298 10302 1
>d 10000 2
10000 : 10146 10292
>d 10146 2
10146 : 10002 10292
>d 10292 2
10292 : 10148 10146
>d 10002 6
10002 : 343 10034 10092 341 10322 0
>d 10148 6
10148 : 344 10180 10239 341 10322 1
>d 10294 12
10294 : -1 10296 10322 10322 0 10300 0 0 1 10304
10304 : 0 0
>t
PC 250 ld r1 1102        r0:0 r1:1 r2:10322 r3:10296 r4:1 r5:0 r6:0 r7:0 r8:0 r9:0
                    r27:341 r28:10322 r29:10034 r30:10092 r31:0
>g
interrupt0
>d 1108 1
1108 : 2
>r

r0 :0    r1 :0    r2 :10322 r3 :10300 r4 :1    r5 :0    r6 :0    r7 :0
r8 :0    r9 :0    r10:0    r11:0    r12:0    r13:0    r14:0    r15:0
r16:0    r17:0    r18:0    r19:0    r20:343  r21:10148 r22:0    r23:0
```

```
r24:0    r25:0    r26:0    r27:341  r28:10322 r29:10180 r30:10239 r31:0
PC:250 RetAds:0 timer0 38 timer1 88 clock 1688
>d 1100 9
1100 : 10000 10292 1 10148 0 10294 10298 10302 2
>d 10000 2
10000 : 10146 10292
>d 10146 2
10146 : 10002 10292
>d 10292 2
10292 : 10148 10146
>d 10002 6
10002 : 343 10034 10092 341 10322 1
>d 10148 6
10148 : 343 10180 10239 341 10322 0
>d 10294 12
10294 : 0 10296 0 0 -1 10300 10322 10322 1 10304
10304 : 0 0
>t
PC 250 ld r1 1102       r0:0 r1:1 r2:10322 r3:10300 r4:1 r5:0 r6:0 r7:0 r8:0 r9:0
                    r27:341 r28:10322 r29:10180 r30:10239 r31:0
>g
interrupt0
>t
PC 250 ld r1 1102       r0:0 r1:1 r2:10322 r3:10296 r4:1 r5:0 r6:0 r7:0 r8:0 r9:0
                    r27:341 r28:10322 r29:10034 r30:10092 r31:0
>g
+2 interrupt0
>d 1108 1
1108 : 2
>r
```

r0 :0 r1 :0 r2 :10322 r3 :10296 r4 :1 r5 :0 r6 :0 r7 :0

r8 :0 r9 :0 r10:0 r11:0 r12:0 r13:0 r14:0 r15:0

r16:0 r17:0 r18:0 r19:0 r20:343 r21:10002 r22:0 r23:0

r24:0 r25:0 r26:0 r27:341 r28:10322 r29:10034 r30:10092 r31:0

PC:250 RetAds:0 timer0 89 timer1 139 clock 2139

>d 1100 9

1100 : 10000 10146 1 10002 0 10294 10298 10302 2

>d 10000 2

10000 : 10146 10292

>d 10146 2

10146 : 10002 10292

>d 10292 2

10292 : 10148 10146

>d 10002 6

10002 : 343 10034 10092 341 10322 0

>d 10148 6

10148 : 343 10180 10239 341 10322 1

>d 10294 12

10294 : -1 10296 10322 10322 0 10300 0 0 1 10304

10304 : 0 0

>

STEP 1

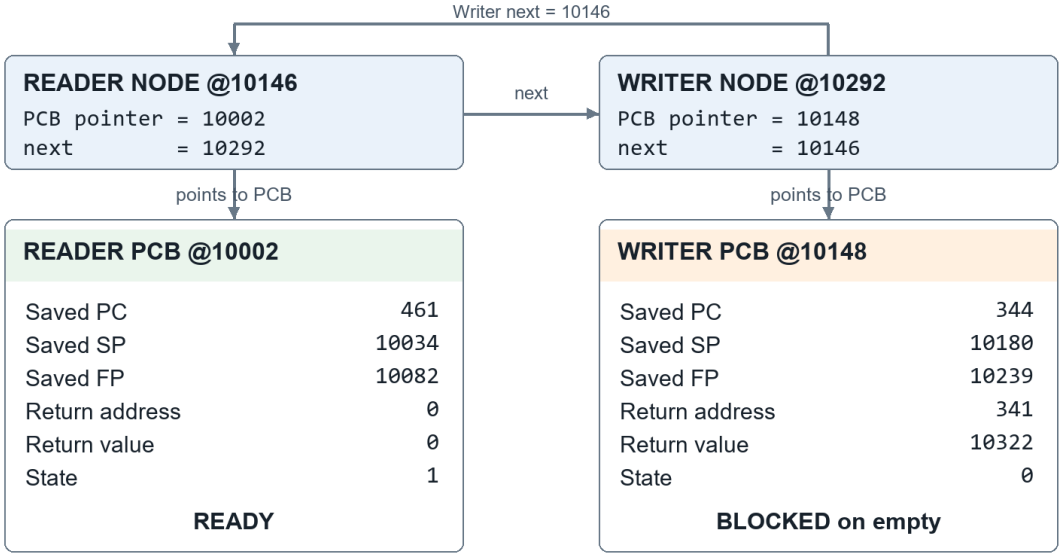
Writer produces 1, then waits

The reader has not consumed 1, so the writer blocks on its next wait(empty).

shareVar = 1 **nump = 1** Snapshot clock: 852

PROCESS QUEUE: the two nodes remain linked in a circle

Q = 10000 | Header: head = 10146, tail = 10292



`qq = 10292 -> Writer node`

`currentp = 10148 -> outgoing Writer PCB`

SEMAPHORES	VALUE	WAITING PROCESS
full	1	None
empty	-1	Writer PCB @10148
mutex	1	None

Next runnable process: READER

Paused at PC = 250, before the scheduler changes currentp.

PCB state: 1 = runnable; 0 = blocked here. All values are decimal.

STEP 2

Reader prints +1, then waits

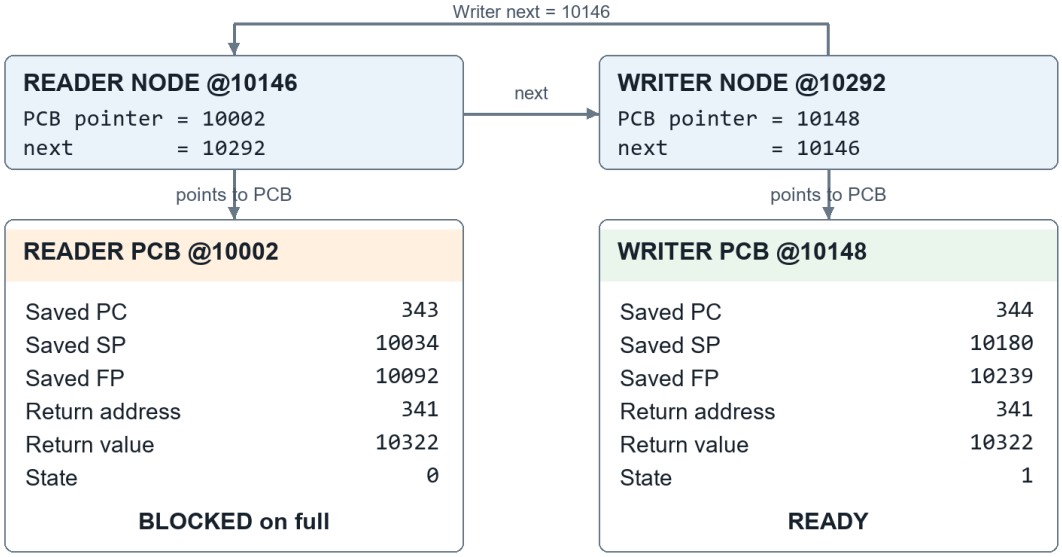
The reader wakes the writer, then blocks on its next wait(full).

shareVar = 1 **nump = 1**

Snapshot clock: 1327

PROCESS QUEUE: the two nodes remain linked in a circle

Q = 10000 | Header: head = 10146, tail = 10292



qq = 10146 -> Reader node

currentp = 10002 -> outgoing Reader PCB

SEMAPHORES	VALUE	WAITING PROCESS
full	-1	Reader PCB @10002
empty	0	None
mutex	1	None

Next runnable process: WRITER

Paused at PC = 250, before the scheduler changes currentp.

PCB state: 1 = runnable; 0 = blocked here. All values are decimal.

STEP 3

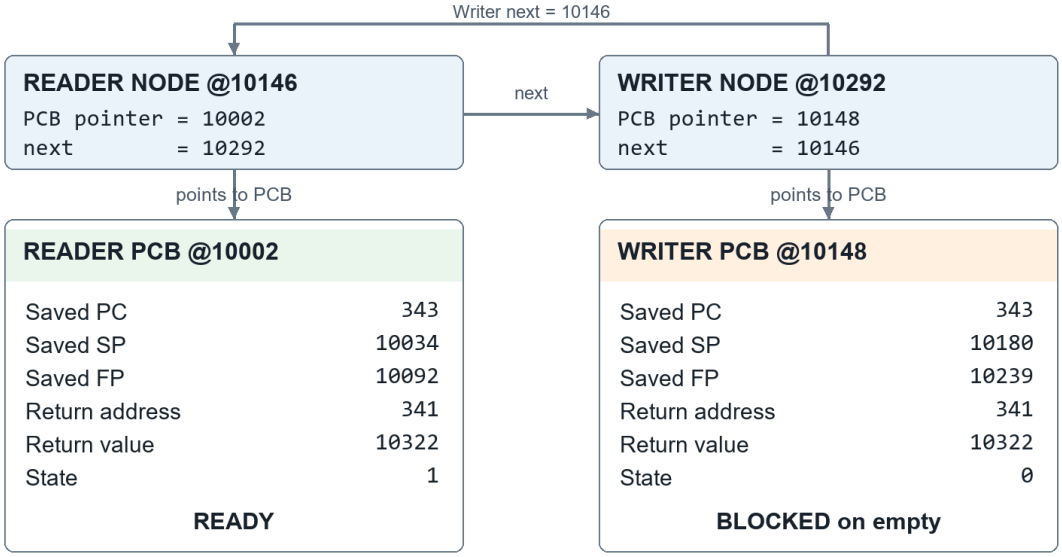
Writer produces 2, then waits

The writer wakes the reader, then blocks on its next wait(empty).

shareVar = 2 **nump = 1** Snapshot clock: 1688

PROCESS QUEUE: the two nodes remain linked in a circle

Q = 10000 | Header: head = 10146, tail = 10292



qq = 10292 -> Writer node

currentp = 10148 -> outgoing Writer PCB

SEMAPHORES	VALUE	WAITING PROCESS
full	0	None
empty	-1	Writer PCB @10148
mutex	1	None

Next runnable process: READER

Paused at PC = 250, before the scheduler changes currentp.

PCB state: 1 = runnable; 0 = blocked here. All values are decimal.

STEP 4

Reader prints +2, then waits

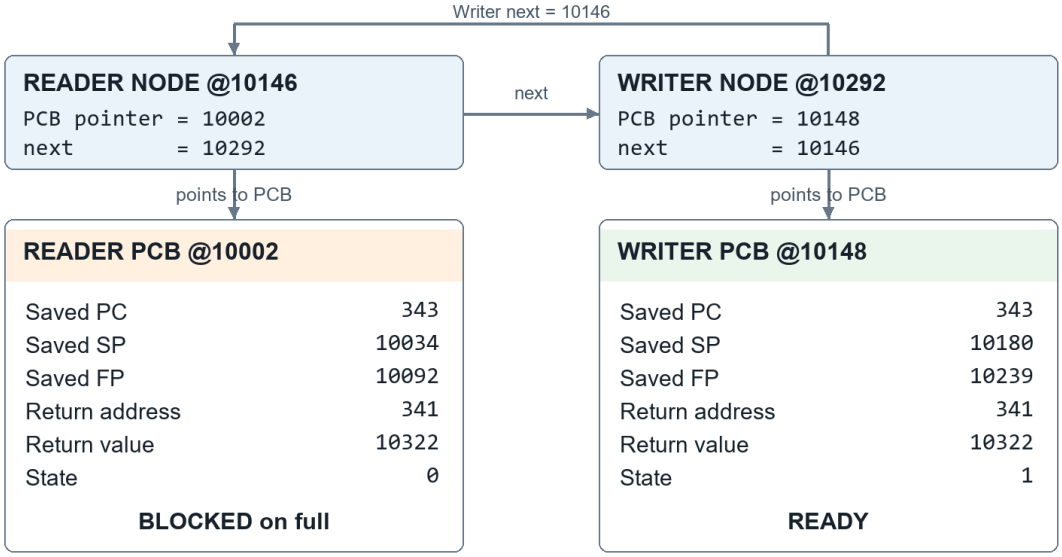
The reader wakes the writer, then blocks on its next wait(full).

shareVar = 2 **nump = 1**

Snapshot clock: 2139

PROCESS QUEUE: the two nodes remain linked in a circle

Q = 10000 | Header: head = 10146, tail = 10292



qq = 10146 -> Reader node

currentp = 10002 -> outgoing Reader PCB

SEMAPHORES	VALUE	WAITING PROCESS
full	-1	Reader PCB @10002
empty	0	None
mutex	1	None

Next runnable process: WRITER

Paused at PC = 250, before the scheduler changes currentp.

PCB state: 1 = runnable; 0 = blocked here. All values are decimal.